

PAP521S PROGRAMMING IN PRACTICE

STATEMENT ABOUT ACADEMIC HONESTY AND INTEGRITY

All staff and students at the Namibia University of Science and Technology (NUST), upon signing their employment contracts and registration forms, commit themselves to adhering to the institution's policies and rules. The core activity of NUST is learning, and in this respect, academic honesty and integrity are very important to ensure that learning is valid, reliable, and credible.

NUST therefore does not condone any form of academic dishonesty, including plagiarism and cheating on tests and assessments, amongst other such practices. NUST requires students to complete their own assignments and produce their own academic work, unless a group assignment is given.

Academic Dishonesty includes, but is not limited to:

- Using the ideas, words, works, or inventions of someone else as if they were your own work.
- Using the direct words of someone else without quotation marks, even if it is referenced.
- Copying from writings (books, articles, webpages, other students' assignments, etc.), published or unpublished, without referencing.
- Syndication of a piece of work, all or part of an assignment, by a group of students, unless the assignment was a legitimate group assignment.
- The borrowing and use of another person's assignment, with or without their knowledge or permission.
- Infringing copyright, including documents copied or cut and pasted from the internet.
- Asking someone else to prepare an assignment for you or to write or sit an assessment for you, whether this is against payment or not.
- Re-submitting work done already for another course or programme as new work, so-called self-plagiarism.
- Bringing notes into an examination or test venue, regardless of whether the notes were used to copy or not.
- Receiving any outside assistance in any form or shape during an examination or test.

All forms of academic dishonesty are considered misconduct under the NUST Student Rules and Regulations. Students found guilty of academic dishonesty will be brought before a Disciplinary Committee and may be suspended from studying for a certain time or expelled. An appropriate endorsement, which will never be erased, is also made on the academic record of ALL guilty students.

COURSE INFORMATION

COURSE CODE AND TITLE:

PAP521S, Programming In Practice

DEPARTMENT:

Software Engineering

PROGRAMME:

Bachelor of Computer Science Honors: Software Development

NOTIONAL HOURS:

Contact hours 30, Self-study hours 50, Assessment hours 20

ASSESSMENT RATIO: This course is Continuous Assessment

Group Project: 40%; Test: 20%; Quizzes / Exercises: 20%; There is no final examination for

this course; Minimum final pass mark: 50%

NQF LEVEL AND CREDIT:

NQF level 10 credits

COURSE DESCRIPTION:

This course aims to give students a deepened understanding of solving computational problems using tools and techniques applied in modern practice. Apart from learning how to use version control systems in managing source code, students will be exposed to writing cleaner code that is self-documenting and tools used for detecting and eliminating inefficiencies. Students also get to select appropriate data structures in teams and use them to develop usable software with multiple levels of abstraction.

PRE-REQUISITES:

Introduction to Programming

COURSE EQUIVALENCIES:

None

COURSE DELIVERY METHODS:

The course will be facilitated through the following learning activities:

- Lectures - this will present the underpinning theories of the course, and students are expected to deepen their knowledge and understanding of the covered subjects through guided self-study.
- Laboratory activities – this will be an avenue for students to apply and acquire practical skills based on the theoretical knowledge they have gained. Laboratory materials will be prepared with a clear indication of what the student is expected to achieve, along with the problem statement and guidelines for approaching the problem.
- Online Quizzes – Allows students to reflect on the concepts learnt in both theory and practical sessions.

The following communication and assessment tools will be used in this course:

MS Teams, E-learning Platform (Moodle), Proctora
(<https://proctor.alphasoftai.com/login>)

Course Format:

- 30 hours of face-to-face (or online via MS Teams) between students and the lecturer.
- 50 hours of out-of-class study time, to be used as follows:
 - 16 hours of laboratory (practical) session
 - 13 hours of reading prescribed material
 - 6 hours of sorting, completing, and annotating own lecture notes
 - 9 hours of working on assignments
 - 6 hours of consultations with the lecturer and fellow students
- 20 hours of assessments:
 - 5 hours of completing online quizzes
 - 5 hours of tests
 - 10 hours of class and project presentations

EFFECTIVE DATE:

Monday 13 July 2026

LECTURER INFORMATION

Lecturer's name: Tendai Mataranyika

Email: tmataranyika@nust.na

Office phone: N/A

Office location: IT House

Office hours: 07h30-12h30

Consultation hours: Tuesdays 17h00

STUDENT READINESS

Technology & Equipment Readiness:

This course will only utilize already existing NUST equipment. Students will be using their own laptops and phones for implementation and must ensure they have an appropriate backup.

Student Commitments and Contact Times:

1. Attend all classes and make appropriate notes. The presentations used in class and laboratory materials will be made available on Moodle. If a student misses a class, it is their responsibility to catch up on what was covered.
2. After every class and before the start of the next class, the student should sort, complete, and annotate their own lecture notes, and select further reading if any particular item was not understood on the first attempt.
3. Students must fully utilize their notational hours to revise, practice, and/or complete homework for the course.
4. Consult the recommended materials to further broaden knowledge and understanding of the subject.

Course Resources:

Presentation slides, laboratory materials, and additional materials will be made available on e-learning (Moodle).

Prescribed Reading:

Sierra, K., & Bats, B., Gee, T. (2022). Head First Java: A Brainfriendly Guide\ (3th ed.). O'Reilly Media. ISBN-10: 1491910771.
ISBN-13: 978-1491910771

Recommended Reading:

Liang, Y. (2019). Introduction to Java Programming and Data Structures, Comprehensive Version, (12th ed.). Pearson, ISBN-10: 0136520235. ISBN-13: 0136520238

STUDENT LEARNING

Learning Outcomes:

Upon completion of the course students will, through assessment activities, show evidence of their ability to:

- Demonstrate a deepened understanding of programming concepts such as input/output, conditions, loops, arrays, strings, and exception handling in developing programs;
- Apply techniques related to data structures and algorithms while writing programs;
- Create methods/functions or procedures, with parameters passed by value and or by reference, and further manipulate these values to be returned or used within the functions or procedures;
- Validate input using regular expressions;
- Debug source code to detect inefficiencies and apply refactoring to proposed solutions;
- Work in teams to produce traceable individual contributions to projects managed using collaborative tool such as Git;
- Develop programs with some Graphical User Interface (GUI);
- Incorporate file processing techniques to read from and write to files within a program;

WEEK-BY-WEEK COURSE SCHEDULE:

Programming in Practice (C Programming)

Course Code: Programming in Practice

Semester: 13 July 2026 – 6 November 2026

Language: ANSI C (C99)

IDE: Visual Studio Code + GCC (MinGW for Windows / GCC on Linux)

Version Control: Git & GitHub

Semester Theme:

Developing a Municipal Financial Management System

Semester Project

Instead of many unrelated programs, students progressively build one application called

Municipal Financial Management System (MFMS)

The system will eventually consist of modules for

- Employee Management
- Payroll
- Assets
- Budget
- Procurement
- Suppliers
- Projects
- Revenue
- Reports

Every week's practical contributes to this system.

Week 1 (13–17 July)

Introduction to Programming in C Theory

- What is Programming?
- Compilation process
- GCC Compiler
- Source code
- Executables
- IDE
- Program Structure
- main()
- Header files

Practical

Write

Municipal Financial Management System

Display

Welcome to Windhoek Municipality

Introduce

```
printf();  
scanf();
```

Week 2 (20–24 July)

Variables and Data Types

Topics

- int
- float
- double
- char
- constants

Lab

Budget Calculator

Input

Revenue

Expenses

Output

Surplus

Deficit

Week 3 (27–31 July)

Operators and Expressions

Topics

Arithmetic

Assignment

Relational

Logical

Increment

Decrement

Lab

Employee Salary Calculator

Input

Basic Salary

Housing

Transport

Tax

Output

Net Salary

Week 4 (3–7 August)

Decision Making

Topics

if

if else

nested if

switch

Lab

Tender Evaluation

Determine

Qualified

Disqualified

Preferred Supplier

Week 5 (10–14 August)

Loops

Topics

for

while

do while

break

continue

Lab

Capture salaries of 50 municipal employees.

Display

Average

Highest

Lowest

Week 6 (17–21 August)

Arrays

Topics

One-dimensional arrays

Searching

Sorting

Lab

Store

Employee salaries

Department budgets

Vehicle registration numbers

Assessment

Practical Test 1

15%

Week 7 (24–28 August)

Strings

Topics

Character arrays

strlen()

strcmp()

strcpy()

strcat()

Lab

Supplier Management

Store

Supplier Name

Email

Phone

Town

Week 8 (31 Aug–4 Sept)

Functions

Topics

Function declaration

Parameters

Return values

Pass by value

Recursion

Lab

Create reusable functions

calculateVAT()

calculateSalary()

calculateBudget()

displayMenu()

searchEmployee()

Week 9 (7–11 September)

Multi-file Programming

Topics

Header files

Separate compilation

Code organisation

Lab

Split system into

employees.c

budget.c

utilities.c

main.c

reports.c

Week 10 (14–18 September)

File Handling

Topics

Text Files

Binary Files

fprintf()

fscanf()

fread()

fwrite()

Lab

Store

Employees

Suppliers

Assets

Budgets

Load them back.

Week 11 (21–25 September)

Structures

Topics

struct

Arrays of structures

Nested structures

Lab

Create

Employee

Supplier

Asset

Budget

Project

Store records.

Assessment

Practical Test 2

15%

Week 12 (28 Sept–2 October)

Pointers

Topics

Memory

Addresses

Pointer variables

Arrays and pointers

Lab

Swap salaries

Pass arrays to functions

Modify employee records

Week 13 (5–9 October) Dynamic Memory Allocation

Topics

malloc()

calloc()

realloc()

free()

Lab

Allow unlimited employees.

Week 14 (12–16 October) Linked Lists (Introduction)

Topics

Nodes

Pointers

Insertion

Deletion

Traversal

Lab

Employee Linked List

Supplier Linked List

Week 15 (19–23 October)
Git and Collaborative Development

Topics

GitHub

Repositories

Commit

Push

Branches

Merge

Each group collaborates on the MFMS project.

Week 16 (26–30 October)
Debugging and Code Quality

Topics

gdb

Breakpoints

Memory leaks

Coding standards

Documentation

Refactoring

Lab

Improve project

Fix bugs

Document code

Week 17 (2–6 November)
Final Project Demonstration

Groups demonstrate

Municipal Financial Management System

Required modules

- Employee Management
- Budget Management
- Supplier Management
- Asset Register
- File Storage
- Search
- Reports
- Menu-driven Interface
- Git Repository

Continuous Assessment

Assessment	Week	Weight
Quiz 1	Week 2	5%
Quiz 2	Week 4	5%
Practical Test 1	Week 6	15%
Quiz 3	Week 8	5%
Practical Test 2	Week 11	15%
Group Project Progress	Weeks 8–16	20%
Final Project Demonstration	Week 17	25%
GitHub Contributions	Throughout	10%
Total		100%

Alignment with the Official Course Outline

This semester plan fully supports the prescribed learning outcomes by covering:

- Input/output using printf() and scanf()
- Conditions, loops, arrays, and strings
- Functions with parameters and return values
- File processing for persistent storage
- Debugging and code refactoring
- Team-based software development using Git
- Data structures, including arrays, structures, pointers, and an introduction to linked lists
- Problem-solving through the development of a medium-sized software application that evolves throughout the semester. These topics are consistent with the course's emphasis on programming fundamentals, algorithms, data structures, file processing, collaborative development, and debugging.

COURSE POLICIES

General Academic Policies:

It is the student's responsibility to be familiar with and adhere to NUST's Policies. These policies can be found in NUST Prospectus or online at www.nust.na/prospectus.

Supplementary Policies:

If a student misses one of the tests for a valid, provable reason, one test will be considered instead of averaging the two. Take note that there are no supplementary assessments for quizzes and projects as the assessments are open for a long period of time.

All late submissions will not be accepted after the due date and time and thus will be marked as 0%.

DATE REVISED: 13 July 2026

FAILURE TO PAY FEES:

A student who fails to pay his/her fees may not be allowed to write the examination and if allowed, the results will be withheld until all outstanding fees are paid in full.

IMPORTANT STUDENT SERVICES AT NUST

There are a variety of services which you can use at the NUST. These services are to your advantage – Use them!!! They include the following:

- Student Counseling and Career Development – Department: Students Services
- Writing Centre and student academic problems –Teaching and Learning Unit (CTL)
- Campus Health and Wellness Centre (CHWC) - Student Services'/ NUST Clinic

AUTHORISATION:

This course is authorized for use by:

Head of Department

13/07/2026
Date

ACKNOWLEDGEMENT BY STUDENT

(To be completed by all students on the course, detached from the course outline and kept on record in the department)

I, _____, hereby acknowledge that I have received this course outline for (insert course title and code), and that I have familiarized myself with its content, in particular the statement about academic honesty and integrity. I agree to abide by the Policies and arrangements spelt out in this course outline.

Signature of Student

Date